



Why Write Unit Tests?

Understanding the importance of automated testing in modern software development

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The Foundation

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Validation Across Scenarios

Unit tests confirm that your code works correctly with various input values

Comprehensive Coverage

They check both **valid** and **invalid parameters** in a consistent, repeatable way

Reliability at Scale

Manual testing becomes unreliable as applications grow in size and complexity

Consistency Guaranteed

Automated tests reduce the chance of forgetting scenarios during repeated testing cycles

Team Collaboration

Especially important when multiple developers contribute code daily

Preventing Future Breakages



Unit tests ensure the **code** you write **continues** to **work correctly** in the **future**

When many developers rely on your method, changes can accidentally break existing functionality

Tests provide a safety net by validating behavior before and after modifications

01

Preserve Expected Outputs

They help guarantee that future updates preserve expected outputs and edge-case handling

02

Long-term Confidence

This creates long-term confidence in the stability of shared methods



Regression Testing

A **regression test** is a type of software test performed to ensure that **new changes in the code (such as bug fixes, new features, or refactoring)** have **not unintentionally broken existing functionality**.

Key points:

- Prevents **reintroduction of old bugs**.
- Ensures **stability** after updates.
- Often automated (e.g., JUnit tests run in CI pipelines).
- Executed whenever the code changes (commits, merges, deployments).

Regression Testing Benefits

- ☐ **Detecting Regressions**
Unit tests are essential for detecting regressions—accidental breaks in previously working features
- ☐ **Early Warning System**
A failing test signals that a new code change has altered expected behavior
- ☐ **Continuous Integrity**
Regression tests ensure continuous integrity as the application evolves
- ☐ **Faster Debugging**
They reduce debugging time by pointing directly to broken functionality
- ☐ **Predictable Development**
Overall, regression testing makes the development process safer and more predictable

Conclusion

Open Floor Discussion

Key Takeaways

- Unit tests provide reliability, repeatability, and long-term confidence in code behavior
- They prevent regressions and reduce the effort needed to test complex features manually
- Automated testing is critical in teams and fast-evolving projects
- Writing tests ensures current and future stability of your application

Discussion Questions

- Have you ever experienced a regression caused by a small code change?
- When maintaining a large codebase, what testing challenges have you faced?
- Do you think manual testing alone is enough for long-term project reliability?

